

UCLA Anderson Math Bootcamp

Charles Rambo

UCLA Anderson

2025

Table of Contents

1 About Me

2 How the Course Works

3 Python

About Me

About Me

Email: charles.tutoring@gmail.com

LinkedIn: [linkedin.com/in/charlesrambo](https://www.linkedin.com/in/charlesrambo)

Please email me your questions at my gmail account. Also, please feel free to connect with me on LinkedIn!

Education

- MFE from UCLA in 2020
- BA in Mathematics from UC Berkeley in 2009
- FRM and CFA certifications

Experience

State Street

Quantitative Analyst, Assistant Vice President (4 months)

- I work in the model risk management team within enterprise risk management
- I use quantitative methods to verify that State Street's credit, liquidity, and macroeconomic models work properly

SSI Investment Management

Portfolio Research Analyst (4 years)

- Firm specializes in convertible securities
- I did valuation, credit, equity, and portfolio construction

Experience

Rambo Tutoring (Self-Employed)

Math Tutor and Author (10 years)

- Did tutoring for precalculus, calculus, linear algebra, differential equations, statistics, probability, and more!
- Wrote study material as well! Check out my stuff on Amazon.com!

How the Course Works

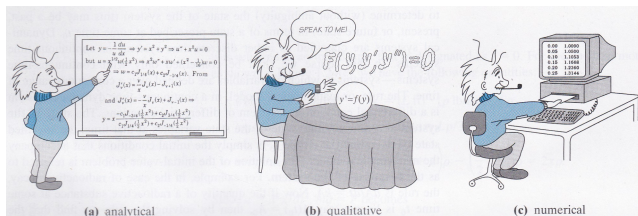
How the Course Works

- Videos available on BruinLearn
- Notes and homework posted on BruinLearn
- Grade of 65% or better to pass; 25% credit given for timely submissions
- No exams

Homework

- Can work in teams of up to four people
- Make sure everybody's name is on the homework when you submit
- Upload homework solutions to BruinLearn
- Homework will be math problems which require some programming
- Submit as an html or pdf file; no screenshots, ipynb, or py files
- I look over the code but I don't run it so more important to make things readable than runnable
- Due dates posted on homework
- Three assignments and one make-up assignment if you missed one or did badly

Approaches to Solve Math Problems



- Work out analytical solutions yourself without a computer
- Very few qualitative problems in this course
- Use Python to obtain numerical solutions

Image from page 28 of *A First Course in Differential Equations* 8th ed. by Zill

Tentative Course Outline

Unit	Description	Sessions
1	Calculus	July 1-15
2	Linear algebra and multivariable calculus	July 17-24
3	Combinatorics, probability, and statistics	July 29-August 7
4	Covariance matrices, PCA, and stochastic calculus	August 12-14

References

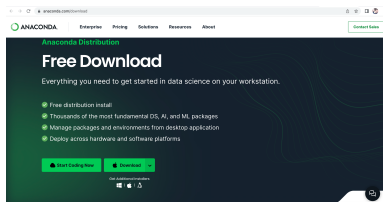
This is an incomplete list of references used to create the notes for this course. You do not need to purchase any of the books listed.

- James Stewart, *Calculus*, Brooke/Cole, 3rd ed., 1995.
- Walter Rudin, *Principles of Mathematical Analysis*, McGraw-Hill, 1976.
- Charles Pugh, *Real Mathematical Analysis*, Springer, 2002
- Serge Lang, *Linear Algebra*, Springer, 3rd ed., 1987.
- Steven Roman, *Advanced Linear Algebra*, Springer, 2nd ed., 2005.
- Morris DeGroot and Mark Schervish, *Probability and Statistics*, Pearson, 4th ed., 2013.
- Marcos Lopez de Prado, *Machine Learning for Asset Managers*, Cambridge University Press, 2020.
- Martin Haugh, *A Brief Introduction to Stochastic Calculus*,
(<https://www.columbia.edu/~mh2078/FoundationsFE/IntroStochCalc.pdf>), Columbia University, 2016.

Python

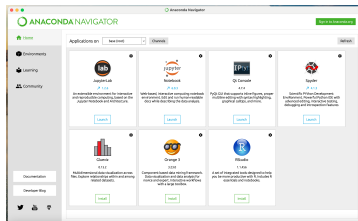
Python Installation

- We'll be using Python to obtain numerical solutions.
- If you haven't used Python before, I suggest downloading Anaconda at <https://www.anaconda.com/download>



Anaconda

After you've installed and opened Anaconda, a screen like this will appear. You have several choices for IDEs. Spyder is useful for data analysis. Jupyter Notebook is popular for explanatory work, so students and teachers tend to use it.



Packages

In this class we'll be using the modules below.

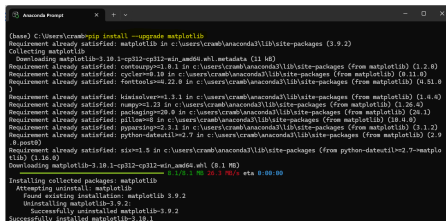
- *NumPy*: Useful for mathematical calculations.
- *Pandas*: Has data frame data type—it allows you to hold data in a format sort of like a spreadsheet.
- *Matplotlib*: Useful for graphing. Somewhat quirky syntax but very popular nonetheless.
- *SciPy*: Useful for scientific computing. It has more advanced math and statistics functions than NumPy.

Package Installation

To install or upgrade a module, go into Anaconda Prompt (Windows) or Terminal (Mac/Linux), and type

```
pip install ...
```

In this example, I'm upgrading matplotlib.



```

Anaconda Prompt
(base) C:\Users\cramb>pip install --upgrade matplotlib
Requirement already satisfied: matplotlib in c:\users\cramb\anaconda3\lib\site-packages (3.9.2)
Collecting matplotlib
  Downloading matplotlib-3.10.1-cp312-win-amd64.whl.metadata (11 kB)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\cramb\anaconda3\lib\site-packages (from matplotlib) (1.2.0)
Requirement already satisfied: cycler<0.10 in c:\users\cramb\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools<=4.22.0 in c:\users\cramb\anaconda3\lib\site-packages (from matplotlib) (4.51.0)
Requirement already satisfied: kiwisolver<=1.3.1 in c:\users\cramb\anaconda3\lib\site-packages (from matplotlib) (1.4.0)
Requirement already satisfied: numpy<=1.23 in c:\users\cramb\anaconda3\lib\site-packages (from matplotlib) (1.26.4)
Requirement already satisfied: packaging<=20.9 in c:\users\cramb\anaconda3\lib\site-packages (from matplotlib) (20.1)
Requirement already satisfied: pillow<=8 in c:\users\cramb\anaconda3\lib\site-packages (from matplotlib) (10.4.0)
Requirement already satisfied: pyparsing<=2.3.1 in c:\users\cramb\anaconda3\lib\site-packages (from matplotlib) (3.1.2)
Requirement already satisfied: python-dateutil<=2.7 in c:\users\cramb\anaconda3\lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six<=1.9 in c:\users\cramb\anaconda3\lib\site-packages (from python-dateutil<=2.7->matplotlib) (1.16.0)
Downloading matplotlib-3.10.1-cp312-cp312-win-amd64.whl (8.1 MB)
  Downloading matplotlib-3.10.1-cp312-cp312-win-amd64.whl (8.1 MB)
Installing collected packages: matplotlib
  Attempting uninstall: matplotlib
    Found existing installation: matplotlib 3.9.2
    Uninstalling matplotlib-3.9.2:
      Successfully uninstalled matplotlib-3.9.2
  Successfully installed matplotlib-3.10.1
```

Python Graphing Example

Example

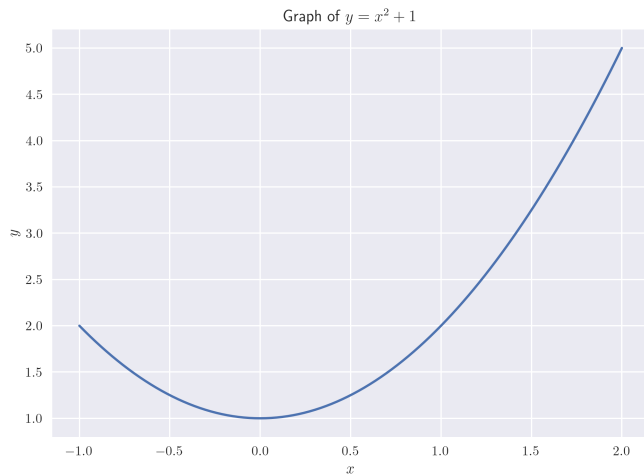
Let $f(x) = x^2 + 1$. Use Python to graph f on the domain $[-1, 2]$.

```
# Import modules
import numpy as np
import matplotlib.pyplot as plt

# These three steps add style
# Use LaTeX
plt.rcParams['text.usetex'] = True
# Increase image resolution
plt.rcParams['figure.dpi'] = 300
# Use Seaborn style
plt.style.use('seaborn-v0.8')
# Ready to do the math part
# Define f
def f(x):
    return x**2 + 1
# Another option is to use a lambda
    function
# f = lambda x: x**2 + 1
# Get 100 x-values in [-1, 2]
x_vals = np.linspace(-1, 2, 100)
```

```
# Use list comprehension to get y-values
y_vals = [f(x) for x in x_vals]
# Automatically vectorized so this works
    too
# y_vals = f(x_vals)
# Generate the plot
plt.plot(x_vals, y_vals)
# Label the x-axis
plt.xlabel('$x$')
# Label the y-axis
plt.ylabel('$y$')
# Give the graph a title
plt.title('Graph of $y = x^2 + 1$')
# Save the figure
plt.savefig(path + 'ex0-1')
# Display the plot
plt.show()
```

Python Graphing Result



Python Optimization Example

Example

Use Python to find the minimum of $f(x) = x^2 + 1$ on the interval $[-1, 2]$.

Solution. From the graph on the previous page, we know that the minimum is $y = 1$ which occurs when $x = 0$. But let's use Python to verify this. Suppose the code above is still in our local environment.

Python Optimization Example

```
# Import minimize from scipy
from scipy.optimize import minimize

# Minimize function; set bounds equal to the domain
minimize(f, x0 = [1], bounds = [(-1, 2)])
```

The output is shown below.

```
message: CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL
success: True
status: 0
    fun: 1.0
       x: [-8.412e-09]
      nit: 2
     jac: [ 0.000e+00]
    nfev: 8
    njev: 4
 hess inv: <1x1 LbfgsInvHessProduct with dtvqe=float64>
```